



UTLA MUKALINGAM

AI SOLUTION ARCHITECT

CONTACT

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 Hyderabad, India
 [LinkedIn Profile](#)
 [GitHub Portfolio](#)
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PERSONAL INFORMATION

Nationality: Indian
Gender: Male
Marital Status: Single
Visa Status: Employment visa required — ready to relocate to UAE/GCC
Driving License: Valid Indian license
Availability: 30 days' notice

CORE EXPERTISE

Agentic AI: Multi-Agent Systems, LangGraph, CrewAI, LangChain, LlamaIndex, MCP, A2A, Tool-Calling, HITL Design, AI Copilots & Intelligent Assistants

LLM & RAG: Claude, GPT-4o, Gemini, Llama, Mistral; Enterprise RAG, Hybrid Retrieval, Re-ranking, Semantic Chunking, Fine-tuning (LoRA/QLoRA), Prompt Engineering

Azure AI: Azure OpenAI, Azure AI Search, AKS, Azure DevOps

LLMOps & Governance: Langfuse, LangSmith, RAGAS, Eval Harnesses, Drift Detection, MLflow; Observability & Security Standards

Vector DBs: pgvector, Pinecone, Weaviate, ChromaDB, FAISS

Voice AI: ElevenLabs, Whisper, Deepgram, Twilio, LiveKit, Sub-800ms Latency

Cloud / DevOps: AWS (EC2, Lambda, SageMaker), Docker, Kubernetes, CI/CD, MLOps, n8n

Backend & Frontend: FastAPI, Django, Node.js, REST Microservices; Next.js 14, React

ML / CV: PyTorch, TensorFlow, Transformers, RF-DETR, YOLO, OpenCV, XGBoost, LSTM

Data: PostgreSQL, Redis, MySQL, Pandas, NumPy

PROFESSIONAL SUMMARY

Senior AI Solution Architect with 7+ years of production-grade experience architecting enterprise-scale Agentic AI and Generative AI solutions. Specialist in multi-agent AI systems (LangGraph, CrewAI, LangChain), enterprise RAG architectures, Conversational and Voice AI, and AI Copilots - delivered cloud-native on Azure (Azure OpenAI, AKS) and AWS with rigorous AI governance, observability, and MLOps. Proven delivery for government and regulated industries across the GCC and India, including Qatar Energy, GCC banking clients, Dr. Reddy's, and the Government of India - plus a live B2B Voice AI SaaS and an enterprise Copilot for Microsoft. Ready to relocate to the UAE/GCC with 30 days' notice.

PROFESSIONAL EXPERIENCE

AI Solution Architect

Jan 2026 – Present

IICL — Intelligence India.Com Limited — enterprise AI solutions company

Hyderabad, India

- › **Architected stateful multi-agent AI systems** using LangGraph & CrewAI with human-in-the-loop gates and deterministic tool-calling, powering procurement and document-intelligence workflows for enterprise clients.
- › **Engineered enterprise RAG architectures** on pgvector with advanced chunking and re-ranking — lifting retrieval precision by 40% and sharply cutting hallucination rates.
- › **Established AI governance & observability standards** — Langfuse tracing, eval harnesses, drift detection, and release gates — applied across all agentic deployments.
- › **Designed AI lead-qualification & appointment automation** integrating Claude API tool-use, RAG, and webhook-driven CRM pipelines — reducing manual sales effort by 50%.
- › **Built iVaak AI** (live at [ivaak.ai](#)) — a multi-tenant Voice AI platform with isolated per-tenant state machines and real-time human escalation; and shipped **TruFix AI**, an autonomous ITSM SaaS (~35% MTTR in pilots).

Lead AI Engineer

Apr 2025 – Jan 2026

LowCode Labs Pvt Ltd — low-code & AI engineering services firm

Hyderabad, India

- › **Deployed enterprise AI agents** (RAG + GenAI) serving 500+ concurrent users at 99.5% uptime via robust MLOps and CI/CD.
- › **Architected multi-agent workflows** on LangChain + FastAPI, cutting deployment time 60% through automated model versioning and container orchestration (Docker/Kubernetes) on AWS.
- › **Built document-processing automation** integrating Claude Vision API and OCR pipelines, plus custom React widgets embedded into enterprise Mendix apps for pharma and FMCG clients.

Lead AI Engineer

Nov 2023 – Mar 2025

The Audience Street Inc. — AI-driven AdTech company

Hyderabad, India

- › **Architected an LLM + NLP AdTech platform** achieving a 25% lift in ad relevance and engagement; integrated GenAI pipelines for ad-copy generation and audience segmentation, accelerating campaign cycles by 40%.
- › **Built prompt-evaluation harnesses** for cost, latency, and regression benchmarking; cut LLM API spend 30% via intelligent caching and batching. Led a cross-functional team of 6.

Senior Data Scientist

Dec 2021 – Nov 2023

Spotflock Technologies — AI/ML products & services company

Hyderabad, India

- › **Deployed a Computer Vision system** (YOLO + OpenCV) for predictive maintenance at Sembcorp — detecting gauge abnormalities in real time, saving 900+ labor hours and \$1M+ in operating costs.
- › **Built a Transformer-based ocean-current model** at 85% accuracy for INCOIS (Ministry of Earth Sciences, Govt. of India), and delivered electricity load forecasting (ARIMA + LSTM) for the Haryana State Government.

Data Scientist

Jul 2019 – Nov 2021

Innodatatics Inc. — data science consultancy

Hyderabad, India

- › **Reduced equipment downtime 30%** with Python predictive-maintenance models across industrial client portfolios; built end-to-end ML pipelines for time-series, NLP, and CV tasks.

EDUCATION

B.Tech, Mechanical Engineering

St. Martin's Engineering College,
Hyderabad · 2015–2019

Prof. Certification — Data Science & AI

360DigiTMG · 2021

CERTIFICATIONS

- IBM — Machine Learning with Python
- AI/ML Certification — UTM (Universiti Teknologi Malaysia)
- AI & Deep Learning — Professional Course
- Data Visualization with Tableau

RECOGNITION

- Empanelled Faculty — Boston Institute of Analytics (GenAI & Agentic AI)
- Speaker — East Coast Maritime & Logistics Summit 2025
- 'Super Scientist' award — Spotflock Technologies
- Medium Writer — LLMs, RAG & Enterprise Agentic AI

LANGUAGES

English — Professional
Hindi · Telugu — Native

FLAGSHIP PROJECTS

Industrial Inspection Automation — BEL (Bharat Electronics Ltd.) 2026 – Present

RF-DETR · RoboFlow · PyTorch · OpenCV · FastAPI

- › Real-time vision-based inspection system detecting bolts, connectors, and assembly components directly from industrial camera feeds on the production line — transformer-based detection (RF-DETR) with automated defect alerts for quality control.

Enterprise AI Copilot — Microsoft (Confidential) 2026

Azure OpenAI · Azure DevOps API · LangGraph · RAG · MS Teams

- › Intelligent assistant embedded in Teams that creates/manages ADO work items, auto-generates test cases, submits pipeline jobs, and answers from an integrated knowledge base — cutting manual ADO overhead and accelerating sprint planning.

iVaak AI — Multi-Tenant Voice AI SaaS (Live) 2025 – Present

LangGraph · ElevenLabs · Whisper · FastAPI · Twilio · Redis · pgvector

- › Production B2B platform giving each tenant an isolated state machine, vector knowledge base, and custom escalation policy at sub-800ms latency; live with active enterprise clients (ivaak.ai).

TruFix AI — Autonomous ITSM Platform 2025 – Present

LangGraph · Claude API · FastAPI · Langfuse · PostgreSQL

- › AI-native layer above Freshdesk/Zendesk; agents own full ticket lifecycle with confidence-threshold human routing — 35% MTTR reduction in pilots, full Langfuse eval before each release.

QConnect — Enterprise AI Training Platform (Qatar Energy) 2025

Next.js 14 · Fastify · FastAPI · pgvector · Claude API · LiveKit · DeepFace

- › Video-learning platform with AI-generated quizzes, multilingual RAG Q&A over port-management manuals, and face-presence monitoring — delivered with formal executive pilot handover.

Pharma Regulatory Document Intelligence — Dr. Reddy's 2025

FastAPI · Claude API · pgvector · LangChain · RAG

- › Enterprise RAG document-intelligence platform for regulatory and compliance documents; semantic chunking with grounded, citation-backed answer generation across large pharma document sets.

AI Interview System 2025

LLMs · FastAPI · React · PostgreSQL

- › End-to-end interview platform integrating LLM-based candidate evaluation with automated scheduling; real-time analysis dashboard providing instant candidate scoring, sentiment analysis, and competency mapping.

AdvertiseWith AI — GenAI Content Platform 2024 – 2025

GPT-4 · Claude · DALL-E · Prompt Engineering · Python

- › Advanced prompt-engineering framework delivering a 40% improvement in generated content quality and relevance, with an intelligent model-routing system automatically selecting the optimal AI model per content requirement.

NinaData — AI-Based AdTech Platform 2023 – 2025

LLMs · NLP · GenAI · Python

- › AI-driven AdTech platform using state-of-the-art NLP and LLMs to analyze and optimize ad content — delivering a 25% improvement in ad relevance and user engagement.

Ocean Current Prediction — INCOIS (Govt. of India) 2022 – 2023

Transformers · Python · AWS · MLOps

- › National-scale spatiotemporal model at 85% accuracy with automated end-to-end MLOps and drift detection for India's national oceanographic service.

Electricity Load Forecasting — Haryana Govt. (HERC) 2022 – 2023

XGBoost · LSTM · ARIMA · Python · Tableau

- › Ensemble forecasting system on Indian Energy Exchange data, deployed for the Haryana Electricity Regulatory Commission to support state-grid resource planning and stability.

